

SKILL-3

```
saketh@DESKTOP-OQ0P5LQ: × + v
GNU nano 7.2 skill3.c *

    buffer[length] = '\0';
    return buffer;
}

/* Main program */
int main() {

    printf("Simple Command History Shell\n");
    printf("Type 'history' to display history\n");
    printf("Type 'exit' to quit\n\n");

    while (1) {

        printf("myshell> ");
        fflush(stdout);

        char *input = read_input();

        if (strlen(input) == 0) {
            free(input);
            continue;
        }

        /* Exit */
        if (strcmp(input, "exit") == 0) {
            free(input);
            break;
        }

        /* Show history */
        if (strcmp(input, "history") == 0) {
            show_history();
            free(input);
            continue;
        }

    }
}
```

```
/* Read input dynamically */
char *read_input() {
    size_t size = INITIAL_SIZE;
    size_t length = 0;

    char *buffer = malloc(size);

    if (buffer == NULL) {
        perror("malloc");
        exit(1);
    }

    int c;

    while ((c = getchar()) != '\n' && c != EOF) {

        /* Prevent buffer overflow */
        if (length + 1 >= size) {
            size *= 2;

            char *temp = realloc(buffer, size);

            if (temp == NULL) {
                free(buffer);
                perror("realloc");
                exit(1);
            }

            buffer = temp;
        }

        buffer[length++] = c;
    }

    buffer[length] = '\0';
}
```

```
saketh@DESKTOP-OQ0P5LQ: × + ▾
GNU nano 7.2 skill3.c *

    if (tail != NULL)
        tail->next = newNode;
    else
        head = newNode;

    tail = newNode;
}

/* Display history */
void show_history() {
    Node *temp = head;
    int i = 1;

    printf("\n--- Command History ---\n");

    while (temp != NULL) {
        printf("%d %s\n", i++, temp->command);
        temp = temp->next;
    }
}

/* Free linked list */
void free_history() {
    Node *temp = head;

    while (temp != NULL) {
        Node *next = temp->next;
        free(temp->command);
        free(temp);
        temp = next;
    }

    head = tail = NULL;
}

/* Read input dynamically */

saketh@DESKTOP-OQ0P5LQ: × + ▾
GNU nano 7.2 skill3.c
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <unistd.h>
#include <termios.h>

#define INITIAL_SIZE 64

typedef struct Node {
    char *command;
    struct Node *next;
    struct Node *prev;
} Node;

Node *head = NULL;
Node *tail = NULL;

/* Add command to history */
void add_history(const char *cmd) {
    Node *newNode = malloc(sizeof(Node));

    if (newNode == NULL) {
        perror("malloc");
        exit(1);
    }

    newNode->command = malloc(strlen(cmd) + 1);

    if (newNode->command == NULL) {
        perror("malloc");
        free(newNode);
        exit(1);
    }

    strcpy(newNode->command, cmd);

    newNode->next = NULL;
```

```
saketh@DESKTOP-0Q0P5LQ:~/skill$ gcc skill3.c -o skill3
saketh@DESKTOP-0Q0P5LQ:~/skill$ ./skill3
Simple Command History Shell
Type 'history' to display history
Type 'exit' to quit
```

```
myshell> ls
Command entered: ls
myshell> fork
Command entered: fork
myshell> mkdir
Command entered: mkdir
myshell> history
```

```
--- Command History ---
1  ls
2  fork
3  mkdir
myshell> exit
Shell terminated successfully.
```